

Mid-Semester Review: Speaker Notes & Talking Points

*Mondrian Conformal Prediction for Uncertainty Quantification in ER Discrete-Event Simulation
Surrogates · Course 23AID201*

Speaking order: Rithvik → Venu → Vipin → Harshith, repeating every 4 slides (13 slides total: 4/3/3/3 split)

Slide 1: Title Slide

Speaker: Rithvik | *On screen: Mondrian Conformal Prediction for Uncertainty Quantification in ER Discrete-Event Simulation Surrogates*

- Welcome everyone - we're presenting our mid-semester review for course 23AID201.
- Our project applies conformal prediction, specifically Mondrian conformal prediction, to a surrogate model trained on an emergency-department discrete-event simulation.
- Quick intros: I'm Rithvik, and with me are Venu, Vipin, and Harshith - we'll each walk through different sections.
- Today's review covers roughly the first half of the project - what's built, validated, and checked so far, and what's planned for the second half.

Slide 2: Contents

Speaker: Venu | *On screen: Roadmap for this presentation*

- Here's the roadmap for the next few minutes.
- We'll start with the problem we're solving and the specific research gap we're addressing.
- Then two short concept slides - one explaining why a marginal coverage number can be misleading, and one showing the Mondrian partition idea - before we get into our actual bridging approach.
- After that: methodology, the real dataset we calibrated on, what's concretely done so far, a snapshot of our literature review, and finally what's planned for end-sem.

Slide 3: Problem Statement

Speaker: Vipin | *On screen: Why point predictions from a surrogate aren't enough*

- Surrogate models are fast, learned stand-ins for expensive simulations, and they're increasingly used to guide real decisions in systems like ER staffing or call-center queueing.
- But a single point prediction isn't enough for a decision like 'will adding two more doctors actually bring the wait time down' - you need a calibrated interval, not just one number.
- Two standard ways to get that interval: Gaussian Processes, which are theoretically solid but expensive to train and rely on distributional assumptions rather than a hard guarantee; and conformal prediction, which is distribution-free and comes with a finite-sample guarantee, but has only been validated for surrogate models in a narrow set of domains so far.

- That's the central question our project is built around: does conformal prediction still hold up when the surrogate is trained on a discrete-event queueing simulation, instead of the physics domains it's already been tested on?

Slide 4: Research Gap

Speaker: Harshith | *On screen: What Gopakumar et al. (2026) leave open*

- Our base paper, Gopakumar et al. 2026, validates conformal prediction for surrogate-model uncertainty quantification, but only in physics domains - PDEs, magnetohydrodynamics, weather, fusion plasma modeling.
- They explicitly flag two open limitations of their own results: first, the guarantee they validate is marginal - averaged over the whole input space, with no guarantee it holds within any specific subgroup or condition. Second, their validation assumes calibration and test data are exchangeable, and they never test what happens under distribution shift.
- Neither limitation has been tested outside physics simulation. A discrete-event, queueing-driven system is structurally different - discrete stochastic arrivals and departures, shared-resource contention, priority queues - not a continuous PDE field.
- That's exactly the gap our project sits in, and it's the reason this domain is a genuinely new test case, not just a repeat of an already-answered question.

Slide 5: Why Marginal Coverage Isn't Enough (concept)

Speaker: Rithvik | *On screen: An illustrative example - not our own results yet*

- This chart is illustrative - a generic example, not our project's own results, since those come in Phase 2. It's here to make one specific concept concrete before we describe our approach.
- Imagine four categories of a system. If you average coverage across all four, you get about 90.5 percent - it looks like your 90 percent target is comfortably met.
- But look closer: Category D alone sits at 68 percent, well below target. A decision-maker who only sees the marginal, averaged number would never know this specific category is unreliable.
- This is exactly the gap Mondrian conformal prediction is designed to close, by calibrating separately within each category instead of pooling everything into one marginal number - which is precisely what our Phase 2 work tests.

Slide 6: Our Bridging Approach

Speaker: Venu | *On screen: Applying Mondrian CP, testing both limitations directly*

- Our core idea: apply conformal prediction, specifically Mondrian CP, to an ER queueing surrogate, and test directly whether Gopakumar et al.'s two limitations actually show up in this domain.
- On the left: addressing marginal coverage. Mondrian CP partitions calibration by category - staffing tercile crossed with arrival-rate tercile - instead of lumping every scenario into one marginal guarantee. The goal is coverage that holds up close to nominal within each category, not just on average.

- On the right: addressing exchangeability. Phase 2 includes a stress test - an out-of-distribution 'surge day' scenario, outside the normal training range.
- That stress test is where we'll actually get to see whether standard CP and Mondrian CP hold up, or break down, once exchangeability is genuinely violated.

Slide 7: The Mondrian Partition (concept)

Speaker: Vipin | *On screen: How the 9-category grid works*

- This is the actual partition structure Mondrian CP will use in Phase 2 - not results yet, just the structure itself.
- Every scenario in our simulation has two covariates: staffing capacity and arrival-rate multiplier. We split each into three terciles - low, medium, high - giving a 3-by-3 grid of 9 categories.
- Every simulated scenario falls into exactly one of these 9 cells, based on its own staffing level and arrival-rate multiplier - nothing is shared across cells.
- Instead of one pooled quantile across all 9 cells, Mondrian CP will calibrate a separate quantile within each one - so a category like the highlighted one, comparatively low-stress with medium staffing and low arrival rate, gets its own calibration, distinct from a high-stress corner like low staffing with high arrival rate.

Slide 8: Methodology Overview

Speaker: Harshith | *On screen: The three-stage pipeline and how each stage is checked*

- Our pipeline has three stages, shown left to right: a discrete-event simulation built in SimPy and calibrated on real ED data; a surrogate model - gradient boosting - trained on the DES's scenario sweep; and uncertainty quantification, comparing a GP baseline against standard CP and Mondrian CP.
- We don't just chain these stages together and hope for the best - each one is checked against the previous one before we move on.
- The DES's output is checked against real aggregated ED statistics - we'll show that number in a moment. Surrogate accuracy is checked against DES outputs held out from training, using MAE, RMSE, and R-squared.
- And critically, all three UQ methods use the identical fixed test split and the identical target coverage of 90 percent, so GP, standard CP, and Mondrian CP end up directly, fairly comparable - not evaluated under different conditions.

Slide 9: Real-World Data Used

Speaker: Rithvik | *On screen: The Kaggle dataset and what's real vs. literature-calibrated*

- We calibrate on the Hospital Triage and Patient History Data from Kaggle - real retrospective data from the Yale New Haven Health System, covering March 2014 through July 2017.

- It's a large dataset: 560,486 ED visits total, 972 variables per visit, across three EDs - one academic and two community sites. We calibrate on Department A only, the largest, academic site: 322,283 visits, about 258 per day.
- We're deliberate about keeping real data and literature-sourced values separate, not blending them silently. From the real data: arrival pattern by 4-hour bin, day of week and month, ESI acuity mix, and department-level daily visit rate.
- From literature: service and treatment time per ESI level, modeled as log-normal - the dataset has no discharge timestamp, and we checked across all 972 columns to be sure before falling back to literature values.

Slide 10: Work Completed So Far (~50%)

Speaker: Venu | *On screen: DES validation, surrogate accuracy, and the GP baseline*

- This slide is our concrete progress checkpoint, roughly half the project.
- First, DES validation: across 200 simulated days, our simulation's mean visits per day is 235.1, against a real value of 258.2 - a 91 percent match, which we consider a strong validation given how much of the calibration is real-data-driven.
- Second, surrogate accuracy: our gradient-boosting surrogate hits an R-squared of 0.929 on total patient count, and even the hardest target - the 95th-percentile wait time - reaches 0.647.
- Third, the GP baseline: at a 90 percent target, it undercovers on three of four targets, landing around 87.7 to 88.8 percent. That's not surprising - GPs rely on distributional assumptions rather than a finite-sample guarantee - and whether conformal prediction, and specifically Mondrian CP, closes that gap is exactly what Phase 2 is for.

Slide 11: The Discrete-Event Simulation

Speaker: Vipin | *On screen: What the DES actually models, and how it's validated*

- This chart shows the real arrival volume by 4-hour bin that our simulation is calibrated on - you can see the clear peak in the 11-to-14 window, which matches typical ED demand patterns.
- Our simulation is a SimPy discrete-event model of a single ED, calibrated directly on this real arrival pattern plus the real ESI acuity mix - not a generic textbook queueing example.
- It uses a priority-resource pool, where higher-acuity patients - lower ESI numbers - get priority for capacity, matching how real triage actually works.
- And as we mentioned, it's validated against real aggregated daily volume at a 91 percent match across 200 simulated days - this simulation is the foundation everything downstream, the surrogate and the uncertainty quantification, is built on.

Slide 12: Literature Review Snapshot

Speaker: Harshith | *On screen: 20 core papers plus 10 real-dataset cross-check papers*

- Our literature review has two parts. The core review covers 20 papers across 5 categories, with 3 reviewed in depth, including Gopakumar et al. 2026, our base paper.

- Those 5 categories: conformal prediction foundations, Mondrian and conditional-coverage CP, surrogate modeling and uncertainty quantification, queueing theory and ED operations research, and discrete-event simulation with ED-specific machine learning.
- On top of that, we have 10 papers that each report real ED service-time or length-of-stay data from their own hospital datasets - we use these to cross-check our own DES's service-time calibration numerically, not just cite them for general support.
- The review also includes a critical assessment section - specifically looking at where the reviewed literature's own assumptions do, or don't, transfer to a discrete-event queueing domain like ours.

Slide 13: Future Scope - End-Sem

Speaker: Rithvik | *On screen: What Phase 2 will deliver*

- To close, here's what's planned for the second half of the project.
- We'll implement standard conformal prediction on the surrogate's residuals, using the DES and surrogate pipeline we've already built and validated in Phase 1.
- Then Mondrian CP itself - partitioning calibration by staffing tercile and arrival-rate tercile, the 9 categories we showed earlier - to measure whether per-category coverage holds up where the pooled, marginal guarantee doesn't.
- We'll stress-test exchangeability with an out-of-distribution demand-surge scenario, run a full comparison of GP, standard CP, and Mondrian CP on coverage and interval width, and bring it all together in the end-sem report: a quantified answer on whether Mondrian CP actually closes the marginal-versus-conditional coverage gap in this domain. Thank you.